Report - Version 1: Agent Base

Introduction

Version 1 is the first working version of the Building AI Agent project.

The goal of this version is to build a simple AI agent from scratch, without using an LLM and without using external frameworks.

This version focuses on the basic flow of an agent:

```
User request
↓
Understand the intent
↓
Extract simple arguments
↓
Run the selected tool
↓
Store the interaction in memory
```

The architecture is intentionally simple because the purpose is to understand the core behavior of an agent before introducing more advanced components.

Goal of Version 1

The main goal of Version 1 is to create a minimal but functional agent.

The agent should be able to:

receive a user request; choose the correct tool; extract simple arguments; execute the selected tool directly; store the interaction in memory.

This version is not meant to be fully professional yet.

It is the foundation of the project.

Architecture

The flow of Version 1 is:

```
User Request
↓
Router
↓
Parser
↓
Agent
↓
Memory
```

In this version, the Agent still executes the selected tool directly.

This means that the Agent has several responsibilities at once.

It coordinates the flow, validates the arguments, executes the function, and stores the result.

For a first version, this is acceptable because the objective is to understand the basic structure.

Tools

The tools are simple Python functions.

Examples:

```
addition(a, b)
subtraction(a, b)
multiplication(a, b)
division(a, b)
power(a, b)
greeting(name)
```

Each tool represents one capability of the agent.

For example:

```
addition(2, 3)
```

returns:

```
5
```

The important concept is that the agent does not solve everything directly.

It selects and uses tools.

Parser

The Parser extracts useful information from the user request.

For example:

```
What is 2 + 3?
```

becomes:

```
{"a": 2, "b": 3}
```

Another example:

```
Hello, my name is luca.
```

becomes:

```
{"name": "luca"}
```

In Version 1, the Parser is simple and rule-based.

It is not generic yet, but it is enough to understand how a user request can be converted into tool arguments.

Router

The Router decides which tool should be used.

It analyzes the user request and looks for keywords or symbols.

For example:

```
What is 2 + 3?
```

is routed to:

```
"addition"
```

While:

```
Hello, my name is luca.
```

is routed to:

```
"greeting"
```

The Router is the first decision-making component of the agent.

It answers the question:

```
Which tool should be used for this request?
```

Agent

The Agent is the main coordinator.

In Version 1, the Agent does the following:

1. 2. 3. 4. 5. 6.

receives the user request; asks the Router to choose a tool; asks the Parser to extract arguments; validates the arguments; executes the selected tool directly; stores the final state in memory.

A typical output state looks like this:

```
{
"request": "What is 2 + 3?",
"action": "addition",
"arguments": {"a": 2, "b": 3},
"result": 5
}
```

This structure is simple but very useful.

It shows the full path from user request to final result.

Memory

Memory is represented as a simple list:

```
memory = []
```

Each interaction is stored as a dictionary.

This allows the agent to keep track of previous requests, selected actions, extracted arguments, and results.

At this stage, memory is very basic.

It does not yet have a dedicated manager.

That will come in a later version.

What Version 1 Teaches

Version 1 teaches the basic logic of an AI agent.

The main idea is:

```
understand → parse → execute → remember
```

This version helps build the mental model of an agent as a system made of multiple steps.

It also introduces the idea that an agent should use tools instead of putting all logic in one place.

Limitations

Version 1 has some natural limitations:

the Agent executes tools directly; the Parser is still rigid; the memory is only a list; there is no Executor component; there is no Planner; there is no Tool Registry; there is no LLM; the agent can only perform one action at a time.

These limitations are expected.

They define the roadmap for the next versions.

Conclusion

Version 1 is the foundation of the Building AI Agent project.

It creates the first working agent and introduces the basic concepts of tools, routing, parsing, execution, and memory.

The most important lesson is that an agent is not just one function.

It is a small system that receives a request, decides what to do, prepares the required data, executes an action, and stores the result.

The next natural step is Version 2, where execution will be moved into a dedicated Executor component.